

Beyond Fake Effective Sample Size: Normalizing Flow Boltzmann Sampling with Initiative Mode Discovery

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1 Introduction

The Boltzmann generator built on normalizing flows (Noé et al., 2019; Wu et al., 2020) is a central tool for sampling from complicated distributions. Let μ denote the target distribution on $\mathbb{R}^d$ and let ν denote the pushforward distribution of the flow. The conventional Boltzmann generator trains the flow by minimizing the reverse KL

$$\text{KL}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx \quad (1)$$

with respect to the flow parameters. On multimodal targets, however, flows trained with the reverse KL are prone to mode collapse: any mode lost during training cannot be recovered, severely degrading sample quality. A standard remedy is to replace the reverse KL with the forward KL (Gabrié et al., 2022):

$$\text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (2)$$

which is more on multimodal targets. The forward KL is the expectation of the *log-ratio* $\log(\mu/\nu)$ under μ , and therefore requires samples from the target. Such samples can be approximated by annealed importance sampling (AIS) (Midgley et al., 2022), which transports the pushforward ν toward μ through a sequence of reweighting and resampling steps. The flow AIS bootstrap (FAB) (Midgley et al., 2022) follows the same idea but trains against the more mode-sensitive weight μ^2/ν in place of μ , enhancing mode discovery.

These objectives can in principle mitigate mode collapse, but their effectiveness hinges on AIS samples that faithfully cover the target. If the pushforward ν misses modes of μ , the AIS chain—initialized at ν —is unlikely to reach them, and the resulting samples remain biased toward the modes ν already covers. This exposes a fatal defect of any AIS-reliant method, including the forward KL and FAB: the effective sample size (ESS) (Doucet et al., 2001), the gold-standard metric for assessing flow quality, can be *fake*. An ESS close to 100% does *not* guarantee mode coverage, since modes missing from ν contribute no mass to the ESS calculation, and their absence goes undetected.

This paper targets the fake ESS problem and develops a training scheme with *initiative mode discovery*, in which prior knowledge of the modes of μ is injected directly into the loss to guide the flow. Doing this safely is far from trivial: the forward KL and FAB are unbiased only when fed *accurate* samples from μ , so naively augmenting the AIS samples with the known mode locations is highly likely to corrupt the minimizer—the injected samples carry no correct importance weights and therefore pull the optimum away from μ .

The center of our training scheme is the cross-regularization functional (denoted X),

$$X_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x, y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (3)$$

where $\omega(x, y)$ is a finite positive measure on $\mathbb{R}^{2d}$ whose choice we defer to a later section. The motivation is direct: since our goal is to match ν with μ , the natural quantity to minimize is the log-ratio $\log(\mu/\nu)$ itself.

We penalize the *difference* of two log-ratios rather than a single absolute value because μ is known only up to an unknown normalization constant, and that constant cancels exactly in the difference. X offers three key benefits as a loss-function ingredient:

1. **Pointwise penalty.** The forward KL reshapes ν globally through Jensen’s inequality, whereas X penalizes log-ratio variations wherever they occur, making it sensitive to local features of the potential landscape that purely global divergences may miss.
2. **Tolerance to inaccurate target samples.** X vanishes whenever the log-ratio is pointwise constant, regardless of the accuracy of the samples drawn from μ . Inaccuracies in the AIS samples therefore barely shift the minimizer, and manually injected mode locations stay safe inside the loss—unlocking the initiative mode discovery promised above.
3. **Drop-in compatibility with the forward KL.** X shares the integrand $\log \nu$ with the forward KL, so when $\omega(x, y) = \mu(x)\mu(y)$ the same autograd graph and the same μ -samples are reused at negligible extra cost; moreover X has the same scale as the forward KL, so no additional tuning is required.

We consider two choices of the measure $\omega(x, y)$ in the X functional (3), each playing a distinct role. The first is $\omega(x, y) = \mu(x)\mu(y)$, giving

$$X_\mu(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x)\mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy. \quad (4)$$

As noted above, computing X_μ and its gradient incurs almost no extra cost beyond the forward KL, so X_μ can be added as a stability- and accuracy-improving regularizer at essentially zero overhead.

The second choice is $\omega(x, y) = \hat{\mu}(x)\hat{\mu}(y)$, sharing the same form as (4):

$$X_{\hat{\mu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \hat{\mu}(x)\hat{\mu}(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (5)$$

where $\hat{\mu}$ need not be an accurate approximation to μ : by the second benefit of X, the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$. The sole role of $\hat{\mu}$ is to place mass on modes that the current pushforward ν has not yet reached, so a coarse mode-finder that errs on the side of broad coverage is exactly what we want. We construct such an $\hat{\mu}$ via the *Quench and Temper* (QT) algorithm, executed in three steps:

1. *Diffusion* (melt): add large Brownian noise to the samples so that they cover a wider region of $\mathbb{R}^d$;
2. *Optimization* (quench): apply LBFGS to drive the noisy samples toward the mode centers of μ ;
3. *Rejuvenation* (temper): apply a short-time Langevin so that the samples relax across the local landscape of each mode.

The three steps are complementary: the Brownian noise enables discovery of modes not yet covered by ν ; LBFGS locates the mode centers precisely; the Langevin step diffuses the samples and makes $\hat{\mu}$ robust to small perturbations. The name borrows from the metallurgical analogue: quenching rapidly cools the material to lock in a low-energy configuration, while tempering gently reheats it to relieve stress and reach a more stable state. The procedure also applies to targets whose mode set forms a low-dimensional manifold, since a finite-time optimization can still capture such a manifold.

Combining the forward KL with both regularizers yields our training loss for initiative mode discovery:

$$\begin{aligned} \mathcal{L}[\nu] &= \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) + \hat{\lambda} X_{\hat{\mu}}(\mu\|\nu) \\ &= \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \iint_{\mathbb{R}^d \times \mathbb{R}^d} \left(\lambda \mu(x)\mu(y) + \hat{\lambda} \hat{\mu}(x)\hat{\mu}(y) \right) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \end{aligned} \quad (6)$$

where $\lambda, \hat{\lambda} \geq 0$ are tunable weights. To evaluate the trained flow we report ESS *together with* the coverage metric (Naeem et al., 2020), which measures how many clusters of $\hat{\mu}$ are reached by samples drawn through ν . The ESS alone can be fake when modes of μ are missing, so it must be paired with a mode-aware diagnostic: only when both ESS and coverage are high does the flow truly approximate μ .

2 Fisher–Rao gradient flow with X functionals

Before we propose the training details with the X functionals, we first study the gradient flow properties of the loss function, whose convergence behavior reflects the difficulty with which the flow can be trained. We view ν as a generic probability density evolving along the gradient flow of the loss under an appropriate Riemannian metric on the space of densities (Ambrosio et al., 2008; Villani, 2009), and ask how fast ν_t converges to the target μ in $\text{KL}(\mu\|\nu_t)$.

Consider the loss function with respect to ν with only the forward KL and X_μ :

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (7)$$

where $\lambda > 0$ is a fixed constant. Throughout this section we abbreviate the importance weight by

$$w(x) := \frac{\mu(x)}{\nu(x)},$$

so that the log-ratio is $\log w(x) = \log \mu(x) - \log \nu(x)$.

The choice of Riemannian metric matters. In the Wasserstein-2 geometry, induced by optimal transport (Jordan et al., 1998; Otto, 2001), the forward KL fails to be displacement convex along Wasserstein-2 geodesics, and the standard Otto–Villani / Bakry–Émery machinery (Otto and Villani, 2000; Bakry et al., 2014) that yields exponential convergence for the reverse KL does not apply. We therefore work in the Fisher–Rao geometry (Amari, 2016; Bauer et al., 2016), in which the forward KL admits an unconditional exponential decay rate and the regularizer X_μ contributes a qualitatively new, nonlocal birth–death modulation (Lu et al., 2019) in which the local update at x is driven by how the importance weight at x ranks against the population of weights under μ .

Denoting the ν -mean by $\langle \cdot \rangle_\nu := \int (\cdot) \nu dx$, the Fisher–Rao gradient of a functional $\mathcal{F}[\nu]$ and the associated dissipation identity along the FR gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t)$ are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \frac{d}{dt} \mathcal{F}[\nu_t] = - \int_{\mathbb{R}^d} \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_{\nu_t} \right)^2 \nu_t dx. \quad (8)$$

The mean-subtraction enforces $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$, so the FR flow conserves probability mass automatically.

2.1 Derivation of X_μ -augmented Fisher–Rao gradient flow

The first variation of the forward KL with respect to ν admits the simple form

$$\frac{\delta \text{KL}(\mu\|\nu)}{\delta \nu}(x) = -\frac{\mu(x)}{\nu(x)} = -w(x), \quad (9)$$

so by (8) the FR gradient of the forward KL is

$$\text{grad}^{\text{FR}} [\text{KL}(\mu\|\nu)](x) = \nu(x) - \mu(x). \quad (10)$$

For the X_μ functional, the inner log-ratio difference has first variation

$$\frac{\delta (\log w(x) - \log w(y))}{\delta \nu(x)} = \frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)}.$$

By the symmetry of X_μ in x and y ,

$$\begin{aligned} \frac{\delta X_\mu(\mu\|\nu)}{\delta \nu}(x) &= 2 \int_{\mathbb{R}^d} \mu(x) \mu(y) \text{sgn}(w(x) - w(y)) \frac{\delta (\log w(x) - \log w(y))}{\delta \nu(x)} dy \\ &= 2 \int_{\mathbb{R}^d} \mu(x) \mu(y) \text{sgn}(w(x) - w(y)) \left(\frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)} \right) dy \\ &= -2 \int_{\mathbb{R}^d} w(x) \text{sgn}(w(x) - w(y)) dy, \end{aligned}$$

where the contribution of $\delta(x - y)$ vanishes because $\text{sgn}(w(x) - w(y)) = 0$ at $y = x$. Hence

$$\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) = -2w(x)S(x), \quad S(x) := \mathbb{E}_{y \sim \mu}[\text{sgn}(w(x) - w(y))]. \quad (11)$$

We call S the *rank field* of the importance weight under μ : $S(x) > 0$ where $w(x)$ exceeds the μ -median of w , i.e. where ν undercovers a mode of μ ; $S(x) < 0$ where ν overcovers; and $S \equiv 0$ at $\nu = \mu$. By antisymmetry,

$$\mathbb{E}_\mu[S] = \mathbb{E}_{x, y \sim \mu}[\text{sgn}(w(x) - w(y))] = 0,$$

and the FR gradient of X_μ with respect to ν is

$$\text{grad}^{\text{FR}}[X_\mu(\mu\|\nu)](x) = -2\mu(x)S(x). \quad (12)$$

Since the FR gradient is linear in the functional, the FR gradient of $\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is the direct sum of (10) and (12):

$$\text{grad}^{\text{FR}}[\mathcal{L}](x) = \mu(x) - \nu(x) + 2\lambda\mu(x)S(x).$$

The augmented Fisher–Rao gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}}[\mathcal{L}]$ therefore reads

$$\partial_t \nu_t(x) = \mu(x) - \nu_t(x) + 2\lambda\mu(x)S_t(x), \quad \text{where } S_t(x) := \mathbb{E}_{y \sim \mu} \left[\text{sgn} \left(\frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right) \right]. \quad (13)$$

Equation (13) is a *nonlocal* PDE: its right-hand side at x depends on $S_t(x)$, which in turn requires the entire profile of the importance weight μ/ν_t over the support of μ . Mass is conserved along the flow because

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x) dx = 2\lambda \int_{\mathbb{R}^d} \mu(x) S_t(x) dx = 0,$$

and the flow is stationary at $\nu_t = \mu$, where $\mu(x)/\nu_t(x) \equiv 1$ and hence $S_t \equiv 0$. When $\lambda = 0$, (13) reduces to the pure Fisher–Rao gradient flow of the forward KL.

2.2 Convergence analysis of the X_μ -augmented flow

To analyze the convergence rate of the augmented flow (13), we take the forward KL $\text{KL}(\mu\|\nu_t)$ as the error metric and differentiate it along the flow:

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = - \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} (1 + 2\lambda S_t) dx + \int_{\mathbb{R}^d} \mu dx = -\chi^2(\mu\|\nu_t) - 2\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx,$$

where the chi-squared divergence is

$$\chi^2(\mu\|\nu) := \int_{\mathbb{R}^d} (w - 1)^2 \nu dx = \int_{\mathbb{R}^d} w^2 \nu dx - 1.$$

The sign of the second integral is settled by the following rank-correlation identity.

Lemma 1 *For any probability density ν on $\mathbb{R}^d$,*

$$\int_{\mathbb{R}^d} \frac{\mu(x)^2}{\nu(x)} S(x) dx = \frac{1}{2} \mathbb{E}_{x, y \sim \mu} [|w(x) - w(y)|] \geq 0,$$

with equality if and only if w is μ -almost-everywhere constant, i.e. $\nu = \mu$.

Proof. By the definition of S ,

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \mathbb{E}_\mu[w(x)S(x)] = \mathbb{E}_{x, y \sim \mu} [w(x) \text{sgn}(w(x) - w(y))].$$

Symmetrizing in $x \leftrightarrow y$ and averaging the two equal copies gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [(w(x) - w(y)) \operatorname{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|].$$

□

Applying Lemma 1 with $\nu = \nu_t$ and $S = S_t$, the dissipation rewrites as

$$\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) = -\chi^2(\mu \| \nu_t) - \lambda \mathbb{E}_{x,y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right]. \quad (14)$$

The elementary inequality $w \log w \leq w^2 - w$ applied to $w = \mu/\nu_t$ gives $\chi^2(\mu \| \nu_t) \geq \operatorname{KL}(\mu \| \nu_t)$, hence $\frac{d}{dt} \operatorname{KL}(\mu \| \nu_t) \leq -\operatorname{KL}(\mu \| \nu_t)$, and Grönwall's lemma yields the exponential decay

$$\operatorname{KL}(\mu \| \nu_t) \leq e^{-t} \operatorname{KL}(\mu \| \nu_0), \quad t \geq 0, \quad (15)$$

with no structural assumption on μ — neither log-concavity, nor a spectral gap, nor any growth condition. This contrasts sharply with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

The uniform convergence rate (15) should *not* be read as a blanket statement that minimizing the forward KL is intrinsically faster or more efficient than minimizing the reverse KL. The FR flow is a continuous-time idealization on the space of densities and is agnostic to the sampling cost. In practice, the difficulty of training has been partially transported to the AIS step: evaluating the forward KL and its gradient requires accurate samples of μ , and producing such samples in high dimensions is itself a hard problem — it is the very reason that AIS is needed, and that the $X_{\tilde{\mu}}$ regularizer must compensate for modes that AIS alone fails to reach. The exponential rate (15) quantifies how fast the flow *can* converge given accurate target samples; it does not eliminate the burden of producing them.

A naive comparison of dissipation rates between KL-only and $\operatorname{KL} + \lambda X_{\mu}$ would also be misleading, since scaling the loss by a constant is mere time reparametrization with no qualitative gain. The substantive contribution of X_{μ} is structural rather than rate-based: while the pure-KL FR gradient is local in x , the rank field S_t is global, so the augmented flow (13) couples the update at x to the entire distribution of importance weights under μ . This nonlocal coupling amplifies the birth rate on undercovered regions of μ and suppresses it on overcovered ones, a global comparison that pure forward KL — which knows only whether $\mu(x)/\nu_t(x)$ exceeds or falls below 1 pointwise — cannot produce.

Remark *The regularizer X_{μ} also decreases monotonically along the augmented flow: a direct computation along (13) gives*

$$\frac{d}{dt} X_{\mu}(\mu \| \nu_t) = -2 \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx - 4\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t^2 dx \leq 0,$$

with strict inequality whenever $\nu_t \neq \mu$. Thus X_{μ} is itself driven to zero in the long-time limit.

2.3 Asymptotic accuracy of biased forward KL training

The convergence analysis of Section 2.2 assumes that the outer expectation in $\operatorname{KL}(\mu \| \nu) + \lambda X_{\mu}(\mu \| \nu)$ is taken under the exact target μ . In practice, the algorithm only has access to an approximation $\tilde{\mu}$ of μ supplied by AIS, and the realized loss reads

$$\tilde{\mathcal{L}}[\nu] = \int_{\mathbb{R}^d} \tilde{\mu}(x) \log w(x) dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) |\log w(x) - \log w(y)| dx dy, \quad (16)$$

where the pointwise ratio $w(x) = \mu(x)/\nu(x)$ is still evaluated against the exact target since it is computed from the target potential U_1 and the flow density per sample. Repeating the derivation of Section 2.1 with μ replaced by $\tilde{\mu}$ in the outer integrands gives the Fisher–Rao gradient

$$\operatorname{grad}^{\operatorname{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\mu}(x)(1 + 2\lambda \tilde{S}(x)), \quad \tilde{S}(x) := \mathbb{E}_{y \sim \tilde{\mu}} [\operatorname{sgn}(w(x) - w(y))].$$

A stationary point ν^* of (16) therefore satisfies the self-consistent equation

$$\nu^*(x) = \tilde{\mu}(x) \left[1 + 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(y) \operatorname{sgn} \left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right) dy \right]. \quad (17)$$

At $\lambda = 0$ the equation collapses to $\nu^* = \tilde{\mu}$: pure forward KL training inherits the bias of the AIS distribution verbatim, and the residual accuracy is exactly $\text{KL}(\mu \parallel \tilde{\mu})$. For $\lambda > 0$, however, the rank-based correction $1 + 2\lambda \tilde{S}^*$ inflates ν^* above $\tilde{\mu}$ where the importance weight μ/ν^* is high (i.e. where ν^* still undercovers μ) and deflates it where the weight is low, providing a global feedback that pulls ν^* from $\tilde{\mu}$ back toward μ . This is quantified in the following theorem.

Theorem 1 *Assume μ is absolutely continuous with respect to $\tilde{\mu}$ and $\lambda \in [0, 1/2]$, so that the bracket in (17) is non-negative and ν^* is a probability density. Any stationary point ν^* of the biased loss (16) satisfies*

$$\text{KL}(\mu \parallel \nu^*) \leq \text{KL}(\mu \parallel \tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (18)$$

Proof. Substituting the stationary identity $\nu^* = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)$ into the difference of forward KLs and using $\mu = \nu^* \cdot (\mu/\nu^*) = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)(\mu/\nu^*)$,

$$\text{KL}(\mu \parallel \tilde{\mu}) - \text{KL}(\mu \parallel \nu^*) = \int_{\mathbb{R}^d} \mu(x) \log(1 + 2\lambda \tilde{S}^*(x)) dx = \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} u(x) \log u(x) dx,$$

where $u(x) := 1 + 2\lambda \tilde{S}^*(x) \geq 1 - 2\lambda \geq 0$ since $\lambda \leq 1/2$. The elementary inequality $u \log u \geq u - 1$ for $u \geq 0$ then yields

$$\text{KL}(\mu \parallel \tilde{\mu}) - \text{KL}(\mu \parallel \nu^*) \geq 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx.$$

Applying the rank-correlation argument of Lemma 1 verbatim with μ replaced by $\tilde{\mu}$ in the outer measure,

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy,$$

which gives (18). $\square$

The right-hand side of (18) contrasts the two roles of X_μ training. At $\lambda = 0$ the bound is tight at $\nu^* = \tilde{\mu}$ and the residual KL is fixed at $\text{KL}(\mu \parallel \tilde{\mu})$ — the AIS bias is a hard floor. Switching on the regularizer with $\lambda \leq 1/2$ strictly lowers this floor by an amount controlled by the discrepancy of the importance weights μ/ν^* across the support of $\tilde{\mu}$: the larger this discrepancy, the more the rank-based feedback pulls ν^* toward μ . The X_μ regularizer is therefore not merely a numerical stabilizer for the forward KL flow: it directly improves the asymptotic accuracy of the trained density when the target samples are biased, which is the practically relevant regime.

The threshold $\lambda \leq 1/2$ is a worst-case condition enforcing positivity of ν^* uniformly in x ; in practice the accuracy improvement persists well beyond it. The practical takeaway is that even without the wide-coverage regularizer $X_{\hat{\mu}}$, adding X_μ alone to the forward KL loss already reduces the bias floor inherited from inaccurate AIS samples.

3 Flow training with X functionals

In this section we derive the total training loss that drives the normalizing flow from the source distribution to the target, combining the forward KL with the X regularizers (4) and (5). Let U_0 and U_1 be the source and target potential functions on $\mathbb{R}^d$, and let $\mu_0 \propto \exp(-U_0)$ and $\mu_1 \propto \exp(-U_1)$ be the corresponding distributions. Given sufficient samples from μ_0 , we seek a differentiable flow map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $G_\# \mu_1 \approx \mu_0$; target samples are then drawn from $G_\#^{-1} \mu_0$ via importance sampling.

The remainder of this section is organized as follows. We first review the forward KL loss and describe how annealed importance sampling (AIS) supplies approximate samples from the target μ_1 . We then detail the computation of X_μ (4), including the autograd-reuse trick noted earlier. Next, we present the Quench and Temper algorithm that constructs the wide-coverage distribution $\hat{\mu}$ used in $X_{\hat{\mu}}$ (5). Finally, we define the ESS and coverage metrics and use a 2D example to demonstrate the power of the X-augmented loss.

3.1 Forward KL and annealed importance sampling

By the change-of-variables formula, the pullback density of μ_0 under G is

$$(G_{\#}^{-1}\mu_0)(y) = \mu_0(G(y)) |\det J_G(y)|, \quad y \in \mathbb{R}^d, \quad (19)$$

where $J_G(y) \in \mathbb{R}^{d \times d}$ is the Jacobian of G at y . The forward KL loss then reads

$$\begin{aligned} \text{KL}(\mu_1 \| G_{\#}^{-1}\mu_0) &= - \int_{\mathbb{R}^d} \mu_1(y) \log(G_{\#}^{-1}\mu_0)(y) dy + \text{const} \\ &= \int_{\mathbb{R}^d} \mu_1(y) \left(U_0(G(y)) - U_1(y) - \log |\det J_G(y)| \right) dy + \text{const}. \end{aligned} \quad (20)$$

We deliberately parameterize the loss in terms of G (mapping target to source) rather than G^{-1} , so that no inversion is required during training.

Evaluating (2) requires samples from μ_1 , which are not directly available. The natural surrogate is to draw $y_1, \dots, y_B$ from the pushforward $G_{\#}^{-1}\mu_0$ and reweight by

$$w(y) = \frac{\mu_1(y)}{(G_{\#}^{-1}\mu_0)(y)} = \exp \left(U_0(G(y)) - U_1(y) - \log |\det J_G(y)| \right), \quad y \in \mathbb{R}^d, \quad (21)$$

followed by a resampling step and a short Langevin simulation to diffuse the duplicates. In high dimensions, however, μ_1 and $G_{\#}^{-1}\mu_0$ typically differ greatly, so the weights $w(y_i)$ collapse onto a few outliers and the resulting samples are of very low quality.

Annealed importance sampling (AIS) bridges this gap by inserting $M + 1$ intermediate distributions

$$\pi_k = (G_{\#}^{-1}\mu_0)^{1-\frac{k}{M}} (\mu_1)^{\frac{k}{M}} = (G_{\#}^{-1}\mu_0) \cdot w(y)^{\frac{k}{M}}, \quad k = 0, 1, \dots, M, \quad (22)$$

so that $\pi_0 = G_{\#}^{-1}\mu_0$ and $\pi_M = \mu_1$. Each rung of the ladder reweights by only $w(y)^{\frac{1}{M}}$, keeping the weight ratio close to 1 and avoiding the degeneracy of direct importance sampling:

Algorithm 1: Annealed importance sampling from $G_{\#}^{-1}\mu_0$ to μ_1

Input: samples $y_1, \dots, y_B$ from $G_{\#}^{-1}\mu_0$, ladder length M

Output: samples $y_1, \dots, y_B$ approximately distributed as μ_1

```

1 for  $k = 1, \dots, M$  do
2   Compute the per-step weights  $w(y_i)^{\frac{1}{M}}$  on  $y_1, \dots, y_B$ ;
3   Resample  $y'_1, \dots, y'_B$  from the weighted empirical distribution;
   // The resampled set has duplicates; Langevin diffuses them
4   Apply a short Langevin simulation to  $y'_1, \dots, y'_B$ , returning fresh samples  $y_1, \dots, y_B \sim \pi_k$ ;
5 end
```

3.2 Single-pass computation of forward KL and X_{μ}

Computing X_{μ} reuses almost all of the forward KL machinery. Substituting the explicit log-ratio under the flow map G (the subscript μ here refers to the target μ_1), the X_{μ} functional (4) becomes

$$\begin{aligned} X_{\mu}(\mu_1 \| G_{\#}^{-1}\mu_0) &= \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu_1(y) \mu_1(y') \left| \left(U_0(G(y)) - U_1(y) - \log |\det J_G(y)| \right) \right. \\ &\quad \left. - \left(U_0(G(y')) - U_1(y') - \log |\det J_G(y')| \right) \right| dy dy'. \end{aligned} \quad (23)$$

Both the forward KL (2) and X_{μ} (23) can therefore be evaluated in a single forward pass that shares the same autograd graph:

Algorithm 2: Single-pass computation of forward KL and X_μ

Input: batch $\mathbf{y} = (y_1, \dots, y_B) \in \mathbb{R}^{B \times d}$ of samples from μ_1
Output: forward KL term and X_μ term
// $\mathbf{z}$ carries the autograd graph through G
1 Compute $\mathbf{z} = U_0(G(\mathbf{y})) - U_1(\mathbf{y}) - \log |\det J_G(\mathbf{y})|$;
// $\mathbf{z}'$ inherits the graph of $\mathbf{z}$ at zero cost
2 Sample a random permutation σ of $\{1, \dots, B\}$ and set $\mathbf{z}' = \mathbf{z}[\sigma]$;
3 Return $\text{mean}(\mathbf{z})$ as the forward KL term and $\text{mean}(|\mathbf{z} - \mathbf{z}'|)$ as the X_μ term;

Consequently, the only autograd work touching the flow parameters is the single backward pass through $\mathbf{z}$, identical to a vanilla forward KL step. Adding X_μ therefore incurs essentially no additional training cost.

3.3 Initiative mode discovery in $X_{\hat{\mu}}$ with QT algorithm

Computing the $X_{\hat{\mu}}$ functional (5) requires samples from $\hat{\mu}_1$, the wide-coverage approximation of the target μ_1 . These samples are prepared once, before training: we generate a fixed pool of P samples $x_1, \dots, x_P$ distributed as $\hat{\mu}_1$, and at each training step we draw a mini-batch of size B from the pool and refresh it with a short Langevin chain before evaluating the loss. The pool is produced by the QT algorithm, which proceeds in three stages: *diffusion* (melt), *optimization* (quench), and *rejuvenation* (temper).

Algorithm 3: QT: initiative mode discovery

Input: samples $x_1, \dots, x_P$ from μ_0
Output: samples $x_1, \dots, x_P$ forming $\hat{\mu}_1$
// Diffusion (melt): scatter the samples to a wider region of $\mathbb{R}^d$
1 Choose a large $\sigma > 0$ and update $x_i \leftarrow x_i + \sigma \xi_i$ with $\xi_i \sim \mathcal{N}(0, I_d)$ i.i.d.;
// Optimization (quench): drive each sample toward a mode center of μ_1
2 Run LBFGS on U_1 for a fixed number of steps to update $x_1, \dots, x_P$;
// Rejuvenation (temper): spread the samples around each mode
3 Run a short Langevin simulation targeting μ_1 on $x_1, \dots, x_P$;

The output samples concentrate near the mode centers of μ_1 , and the initial Brownian diffusion ensures that even modes far from the support of the input are reachable. The $X_{\hat{\mu}}$ regularizer therefore penalizes any mode of μ_1 on which the current pushforward $G_{\#}^{-1} \mu_0$ places too little mass, driving the flow to discover modes that AIS alone could never reach.

A side effect of the diffusion step is that a large σ biases the output samples toward the outer modes—those far from the bulk of μ_0 . This bias is not only harmless but actually desirable for mode discovery, since the modes that the current pushforward fails to cover are precisely those lying far from μ_0 .

3.4 Evaluation metrics: ESS and coverage

The trained flow G induces the distribution $G_{\#}^{-1} \mu_0$ as an approximation to the target μ_1 . We assess its quality through two complementary metrics: the effective sample size (ESS), which measures how uniform the importance weights are on the support reached by the flow, and the coverage, which checks whether every mode of μ_1 has been reached at all.

The first metric is the effective sample size. Let $\mathbf{y} = (y_1, \dots, y_N) \in \mathbb{R}^{N \times d}$ be i.i.d. samples from $G_{\#}^{-1} \mu_0$, with importance weights $w(y_i)$ given by (21). The ESS is defined as

$$\text{ESS}(\mathbf{y}) = \frac{(\sum_{i=1}^N w(y_i))^2}{N \sum_{i=1}^N w(y_i)^2} \in [0, 1], \quad (24)$$

where the upper bound 1 follows from the Cauchy–Schwarz inequality and is attained when all weights are equal. When G is perfectly trained the weights are constant and the ESS reaches 1. As emphasized in the

introduction, however, this conclusion is only one-way: a high ESS certifies that the weights are uniform *on the support of* $G_{\#}^{-1}\mu_0$, but modes of μ_1 that the pushforward never reaches contribute nothing to the sum and so cannot lower the ESS—hence the *fake ESS* pitfall.

To detect such missing modes, we pair the ESS with the coverage metric of [Naeem et al. \(2020\)](#). Let $\mathbf{x} = (x_1, \dots, x_P) \in \mathbb{R}^{P \times d}$ be the samples of $\hat{\mu}_1$ produced by the Quench and Temper algorithm (Algorithm 3). For each $1 \leq i \leq P$, the k -nearest-neighbor radius of x_i is

$$\text{NND}_k(x_i; \mathbf{x}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq P : 0 < |x_i - x_j| < r\} \geq k \right\}, \quad (25)$$

i.e., the distance from x_i to its k -th nearest neighbor in $\mathbf{x}$. The coverage of $\mathbf{x}$ by the pushforward samples $\mathbf{y}$ is then

$$\text{Coverage}_k(\mathbf{y}; \mathbf{x}) = \frac{1}{P} \sum_{i=1}^P \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - x_i| \leq \text{NND}_k(x_i; \mathbf{x}) \right\}, \quad (26)$$

i.e., the fraction of points in $\mathbf{x}$ whose k -nearest-neighbor ball contains at least one pushforward sample. A coverage close to 1 certifies that every region populated by $\hat{\mu}_1$ has been reached by the flow, while a coverage substantially below 1 exposes precisely the missing modes that the ESS alone cannot see.

4 Construction of adaptive Boltzmann generator

We state how to construct a adaptive Boltzmann generator which allows to determine the temperature ladder adaptively and derive a numerically stable training scheme. In general, for a d -dimensional system, an empirical choice of the ladder length in AIS (Algorithm 3) is $M = d$. We can use a simple Gaussian example to show this:

5 Numerical experiments

5.1 Single-step flow in low-dimensions

5.1.1 2D benchmark distributions

We test the same training scheme on two complementary 2D targets, Himmelblau and Annulus, listed in Table 1. Himmelblau places four isolated modes at the corners of a square; Annulus is rotationally symmetric, with a thin one-dimensional ring of minima at $r = 2$ together with an isolated minimum at $r = 0$.

name	potential $U_1(x_1, x_2)$	minima
Himmelblau	$(x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$	four isolated points
Annulus	$20(r^6 - 8r^4 + 16r^2 + 1)^{1/3}, r^2 = x_1^2 + x_2^2$	$r = 0$ and ring at $r = 2$

Table 1: Two 2D benchmark targets.

For both targets the source μ_0 is an isotropic Gaussian $\mathcal{N}(0, \sigma^2 I)$ ($\sigma = 1.0$ for Himmelblau, $\sigma = 0.8$ for Annulus). The flow is a neural spline flow with 6 coupling transforms, 32 bins, hidden widths (128, 128), on a square box $[-6, 6]^2$ for Himmelblau, $[-3.6, 3.6]^2$ for Annulus). Training uses Adam with learning rate 10^{-3} , batch size 500, for 1000 steps; at each step the AIS surrogate is built by one importance-weighted resampling pass against the current pushforward, followed by 50 Langevin steps of size 2×10^{-3} . We fix $\lambda = \hat{\lambda} = 1$ throughout. The wide-coverage measure $\hat{\mu}_1$ is built once at the start of training by applying QT (Algorithm 3) to a fresh batch of 500 source samples, with diffusion standard deviation 2, optimization step 0.5 for 200 LBFGS iterations, and rejuvenation step 2×10^{-3} for 1000 Langevin iterations; the resulting samples are recycled (with one Langevin rejuvenation per training step) inside the training loop.

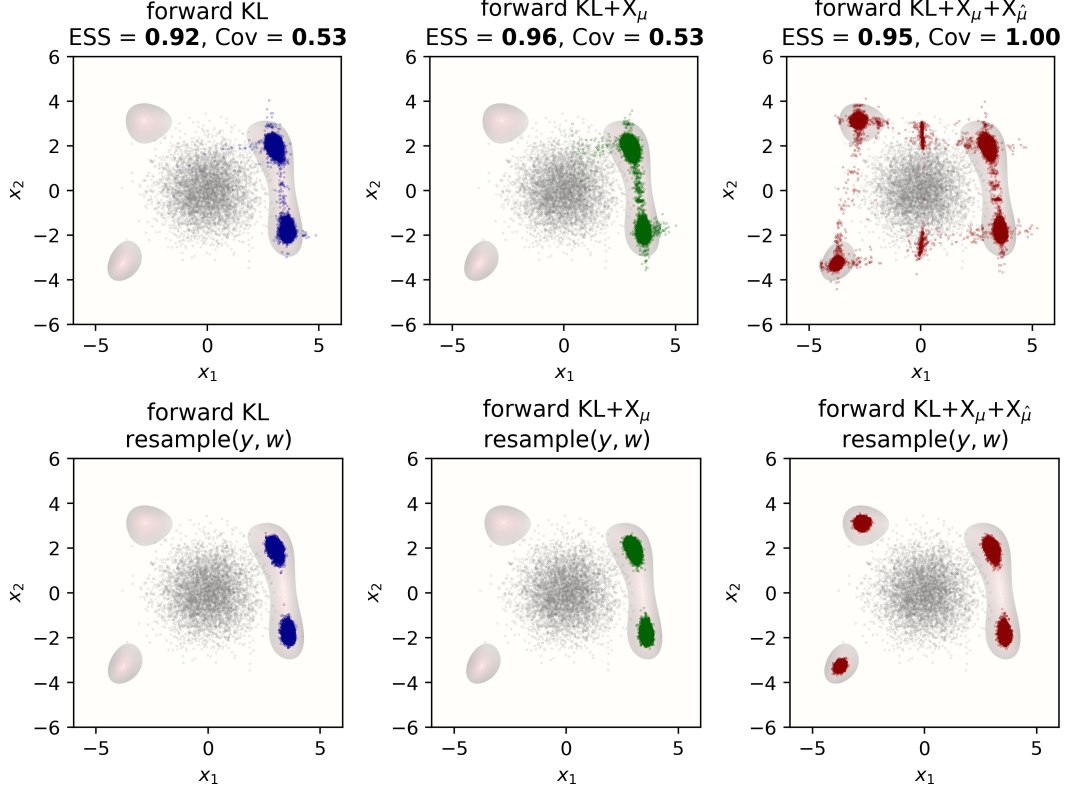


Figure 1: Himmelblau target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns: forward KL, forward KL + X_μ , forward KL + $X_\mu + X_{\hat{\mu}}$. Each pushforward panel title reports the final ESS and the $k = 5$ coverage of the QT-generated $\hat{\mu}_1$.

Himmelblau (mode-coverage regime). The source places nearly all its mass near the central saddle, so the initial pushforward is far from any of the four Himmelblau modes. Both forward KL and forward KL + X_μ converge to a flow that covers only two of the four modes (coverage ≈ 0.53 in both cases), and the ESS alone gives no warning that the other two modes have been missed. Adding $X_{\hat{\mu}}$ steers mass toward the QT-discovered modes and the trained flow reaches coverage 1.00 (Figure 1). This confirms the role of $X_{\hat{\mu}}$ as the safeguard one needs precisely when μ_0 does not by itself cover all modes of μ_1 — the regime in which the forward KL alone is silent.

Annulus (shape-correction regime). All three methods reach $k = 5$ coverage 1.00, so $X_{\hat{\mu}}$ contributes no mode rescue here; the discriminating quantities are the flow shape and ESS. Forward KL alone settles at $\text{ESS} \approx 0.66$, leaving visible axis-aligned filaments across the central barrier. Adding X_μ tightens the pushforward to a clean ring-plus-point geometry and lifts ESS to ≈ 0.95 (Figure 2); appending $X_{\hat{\mu}}$ on top changes neither qualitatively. This case isolates the role of X_μ as a shape regularizer: it pointwise penalizes log-ratio variations that the global forward KL leaves untouched.

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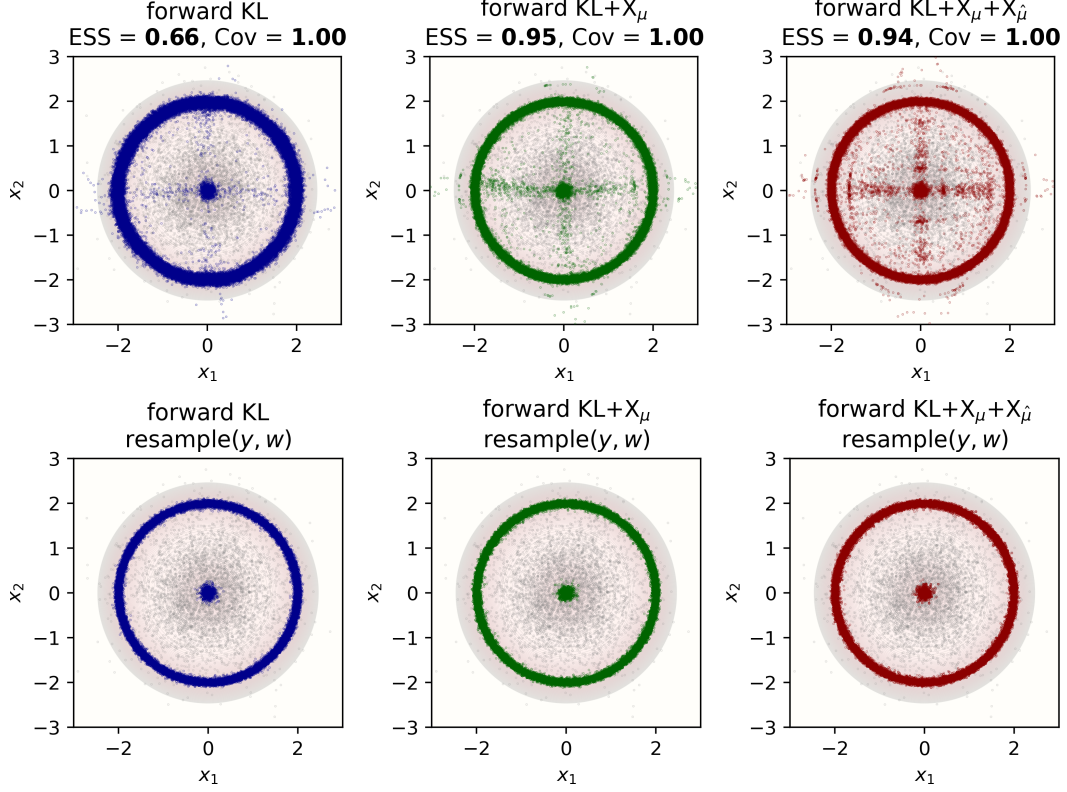


Figure 2: Annulus target. Row 1: pushforward $y = G^{-1}(x)$. Row 2: $\text{resample}(y, w)$ with Langevin rejuvenation. Columns: forward KL, forward KL + X_{μ} , forward KL + $X_{\mu} + X_{\hat{\mu}}$. Each pushforward panel title reports the final ESS and the $k = 5$ coverage of the QT-generated $\hat{\mu}_1$.

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